Neural means and kernel corrections for an EMIT atmospheric radiative-transfer emulator

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Abstract

Imaging-spectroscopy retrievals call an atmospheric radiative-transfer model many times per spectrum, and at mission data rates learned emulators of the tabulated physics are a practical necessity. We construct and compare emulators of a tabulated radiative-transfer model on the 285-band EMIT wavelength grid, mapping a six-dimensional atmospheric and viewing state to four spectral components — path radiance, direct flux, diffuse flux, and spherical albedo — from which radiance and surface reflectance are derived. Every model is trained, selected and scored under a fresh-seed protocol: ten random splits, every choice made on each split's own validation rows, one test read per family, and the numbers reported as means and standard deviations over the splits. An exact Matérn regression fitted on all 18,884 training rows with one length scale per input reaches 0.095% relative radiance error at reflectance 0.7, against 0.76% for a cubic polynomial and 0.39% for a fully connected network; its learned metric recovers the physics of the components, giving the relative azimuth a sixteen-fold longer length scale on the flux and albedo maps at every split. A kernel regression of the network's residuals brings the network to 0.14%, a kernel on the network's last-layer features to 0.098%, and a per-component convex stack of the heads, fitted on validation, to 0.087%, the best row of the table but by less than the spread across splits; per-coordinate winner-take-all selection does not transfer from validation to test. The residual correlations between the kernel and the network ensemble satisfy the classical admission condition for combination on every component, and the spherical albedo remains the hard component for every family. We also quantify an irreducible floor in the reflectance metric: within the deep water-vapor absorption windows the transmitted flux approaches zero and the inverse sensitivity is unbounded; at exactly zero flux, reflectance is non-identifiable, while finite-precision cancellation near that singularity produces a measured floor even when the exact tabulated components are supplied. Applied without structural change to the OCO-2 reduced-radiance problem of the companion paper, under the same ten-split protocol, the construction that spends the kernel's exactness inside the network's representation is again the one that pays, and a control isolates it: an exact Matérn regression on a network's frozen features beats a ridge readout refitted on the same features at every split, on every band and under every training loss, while the ridge readout does not beat the network it reads out from in the metric that network was trained for. Three networks trained under losses that weight the output coefficients differently divide the two reported metrics between them, and because the reconstruction basis is orthogonal a per-coefficient choice made on validation takes both: it assigns the leading coefficient to the kernel head of the weighted-coefficient network at every split and the rest to that of the coefficient-trained one, and it is more accurate than

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the release's stored kernel-flow emulator on both metrics on all three bands. On the member predictions of both campaigns we then ask which combination rule is right. Random-matrix estimators that clean the covariance of the predictions lose; a minimum-variance stack on the covariance of the errors, in the reported metric and with nonnegative weights, has no tuned constant, gains 4.6% over the best cross-validated heuristic on EMIT and loses 1.3% on OCO-2, and a ladder of covariance cleaners shows that the constraint, not the cleaning, is the estimator; a label-free choice between the constrained and least-squares stacks removes the corpus dependence. A reevaluation on the release's 976 retrieval records, against the physics model's own radiances, gives the reference's per-coordinate Gaussian processes roughly 2.1 to 5.3 times lower error than the budget-matched networks of the earlier single-split study. An identity check changes the interpretation: all 976 reduced states match stored test inputs within 6.67×10^{-16} (974 distinct test rows, with the same nearest row in all three bands). This is therefore a comparison against a different radiance target at test-associated states, not evidence from unseen states or from an independent third pool.

1 Introduction

Imaging spectroscopy missions such as EMIT [13], SBG, and AVIRIS-NG invert measured radiances for surface and atmospheric quantities, and the inversion loop calls an atmospheric radiative-transfer model many times per spectrum. At mission data rates the physics code itself (MODTRAN in the pipelines relevant here [1]) is far too slow, so operational retrievals interpolate precomputed lookup tables, and a body of work replaces table and physics alike with learned emulators: Gaussian-process surrogates [5], the neural-network MODTRAN emulator that serves the EMIT mission's operational atmospheric correction [3, 2], and, recently, physics-guided and multi-fidelity architectures for correction coefficients [4]; network training itself can draw on a growing toolbox of refinements [7, 8]. Both families have a record on emulation tasks of this kind: networks learn their own representations and scale easily to wide spectral outputs [10, 11], while kernel and Gaussian-process methods come with exact solves and a strong record both on operator-learning benchmarks [9] and on smooth, low-dimensional regression [14]. Which family is preferable for a given emulation problem, and whether they can be combined so that neither's advantage is given up, are practical questions that the structure of the data itself should decide.

This paper studies those questions on a tabulated radiative-transfer emulation problem over the EMIT wavelength grid, using the coupling construction of [6]: train a neural network as the mean in the metric the application reports, correct it with exact Matérn kernel regressions — of its residuals on the input space, and of the target on the network's own learned features — and let a per-coordinate selection on held-out data choose, coefficient by coefficient, among the network, the kernels, and the corrected predictors. We ask what structure the dataset has, which model families that structure favors, and whether the kernel correction of a neural mean improves on both of its parents here.

The dataset answers the first question quickly (Section 2). The input is six-dimensional, and two coordinates carry most of the variance of the output; the output spectra are smooth curves whose empirical rank is a few dozen out of 285 bands; and the map from state to spectrum is smooth enough that cubic polynomial regression already lands near two percent relative error. In the taxonomy of [6] this is the regime that favors exact kernel methods, in contrast to emulation problems with higher-dimensional reduced states where representation learning carries more of the weight. The interest of the comparison is therefore not whether some exotic architecture wins, but how the familiar families divide the problem between them when the geometry is easy, and whether the coupling machinery preserves the best of each.

Sections 3 and 4 give the protocol and the measurements. In brief: with every training row in the solve and a length scale per input, the exact kernel is the strongest single model on this table and its learned metric recovers the physics of the components; the exact kernel regression of the network's residuals does what it was built to do, handing the network the kernel's accuracy on the smooth components, but it is no longer the lowest row; a per-component convex stack of the heads is, by about one standard deviation across the ten splits, while per-coordinate selection does not transfer from validation to test; and a fixed floor in the reflectance metric is the signature of atmospheric absorption windows in which the transmitted flux approaches zero, making the inverse sensitivity unbounded, and at exactly zero leaves the physical reflectance non-identifiable. We quantify that floor and summarize reflectance through statistics the divergence cannot dominate. Section 5 repeats the construction on the OCO-2 problem of the companion paper under the same protocol, where representation rather than smoothness is the lever; Section 6 takes the member predictions of both campaigns and asks which rule for combining heads is the right one, and why; and Section 7 tests the reading on seven further emulation corpora under one protocol.

2 Data and protocol

The data are $n = 23{,}313$ tabulated radiative-transfer evaluations. Each input $x \in \mathbb{R}^6$ collects the atmospheric and viewing state; the coordinate ranges are consistent with the standard lookup-table axes for imaging-spectroscopy atmospheric correction [2] — an aerosol optical depth in $[0.05, 1.2]$, a surface elevation in $[-0.5, 6]$ km, a water-vapor column in $[0.05, 7]$ cm, a relative azimuth in $[0, 180]^\circ$, a solar zenith angle in $[0, 65]^\circ$, and a view zenith angle within 30° of nadir. Each output is a set of four spectral components on the same 285-point EMIT wavelength grid spanning 381–2493 nm: path radiance Y_1, direct flux Y_2, diffuse flux Y_3, and spherical albedo Y_4 (Figure 1). The quantities retrieved downstream are the radiance at a test surface reflectance ρ,

$$L(\rho) \;=\; Y_1 + \rho\,\frac{Y_2 + Y_3}{1 - \rho\,Y_4}, \tag{1}$$

and the reflectance recovered by inverting (1) with predicted components at the true radiance,

$$\hat{\rho} \;=\; \frac{L - \hat{Y}_1}{\hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)}, \tag{2}$$

both evaluated at $\rho = 0.7$.

The protocol is the same for every family. The table is split at random into a test block of 2,331 evaluations and a training block of 20,982, and a further ten percent of the training block, 2,098 points, is set aside for validation, leaving 18,884 training points. The split is drawn ten times, from ten seeds, and every model is trained, selected and scored once per seed: inputs and outputs are standardized on the training rows of that seed, every hyperparameter and every combination weight is chosen on that seed's validation points, and the test block of that seed is read once per family at the end. The numbers reported below are means and standard deviations over the ten seeds, so a difference between two rows can be read against the spread of each. An earlier version of this study used a single split and consulted its test block during exploration; none of its numbers is reused here, and the ten seeds were drawn afresh, by a plain permutation rather than the earlier library routine, so no code path shares randomness with that split. The

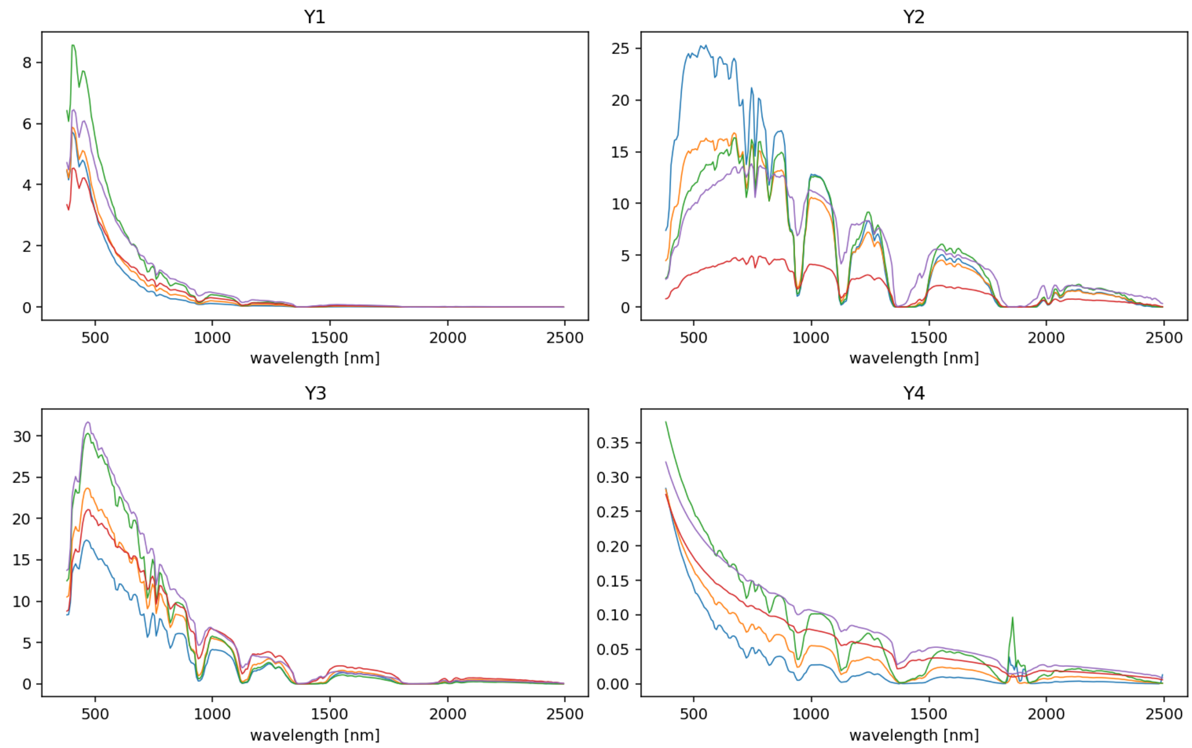

Figure 1: Example output curves across the EMIT wavelength grid, five samples per panel. The curves are smooth, share the deep water-vapor absorption features near 1380 and 1900 nm, and differ mainly by smooth rescalings along the state axes.

validation rows are used for early stopping, checkpoint selection, hyperparameter selection and the combiners, and are therefore not an independent calibration split; no coverage guarantee is claimed from them.

Three structural facts, measured before any model was chosen, organize everything that follows.

The outputs are low rank. On standardized training outputs, 99.99% of the variance of every component lies in the first 11–25 principal components, and 99.9999% in at most 44 (Figure 2, left); the minimum adjacent-band correlation across the four components is 0.994952 on the training rows. All regressors below therefore act on the first 64 principal coefficients — a lossless reduction at the 10^{-6} level, verified by round trip — except the networks, which we also train directly on the 285 standardized bands.

The measured input relevance is strongly anisotropic. Shuffling the aerosol coordinate destroys a linear model's validation R^2 (a drop of 0.89); the solar zenith follows at 0.54; elevation and water vapor sit near 0.1; the two remaining angles barely register (Figure 2, right). The values reported for these probes, and for the other structure measurements used to motivate architectures, are their validation-split rescoring; the initial exploratory versions were test-scored as disclosed above. Six nominal dimensions have four coordinates with measurable first-order relevance, two of them dominant. This finite-sample diagnostic helps explain why smooth methods work, but a selected scalar length scale does not remove anisotropy. Any theorem attributing generalization to anisotropic metric adaptation would also have to control regularization bias,

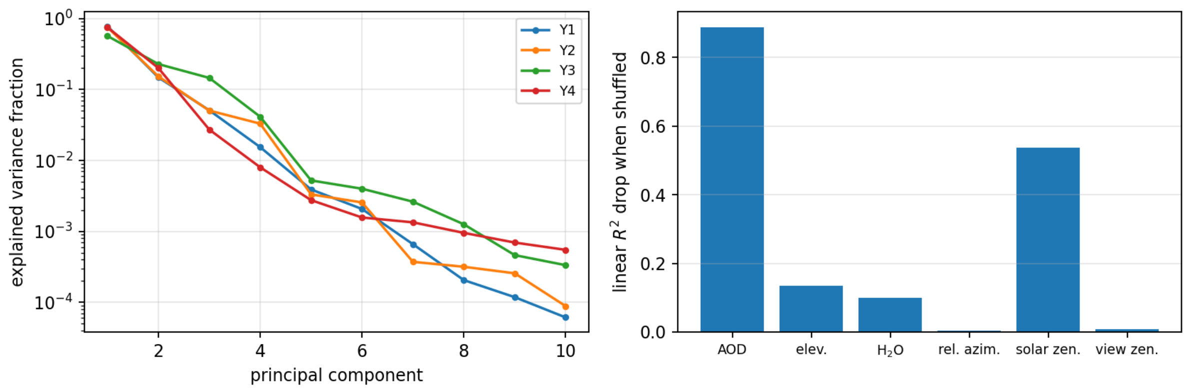

Figure 2: Left: explained-variance spectra of the standardized outputs. Every component concentrates 99.99% of its variance in 11–25 principal components out of 285 bands. Right: drop in linear validation R^2 when a single standardized input coordinate is shuffled. Aerosol optical depth and solar zenith angle dominate; the azimuth and view-zenith coordinates are almost inert at first order.

selection of scale and nugget, and the empirical output-rank truncation. We use anisotropy as a measured property of this split, not as such a guarantee.

The map is smooth. A linear model in the standardized state reaches $R^2 = 0.83$; distance-weighted 5-nearest-neighbors does worse (0.74), which already says local averaging is the wrong bias here; and a cubic polynomial ridge reaches mean relative L^2 errors near two percent (Table 1). Smooth, low-dimensional, low-rank: in the taxonomy of the companion paper this is the regime of the Helmholtz and Navier–Stokes benchmarks, where the kernel stage won the internal selection, not the regime of its OCO-2 problem, where representation learning was the lever.

One more property of the task matters for reading any reflectance number. In the deep absorption windows the transmitted flux $Y_2 + Y_3$ collapses toward zero — in this table, to values as small as 10^{-21} — and the inversion (2) is undefined at a fully opaque band (0/0): the physical reflectance is non-identifiable there. The protocol maps a non-finite inverse to zero, but that branch is not what creates the measured true-component floor. Feeding the *true* components through the evaluation yields a reflectance RMSE of 0.0054 against the true $\rho = 0.7$, produced entirely by 0.25% of (sample, band) entries — 1,652 of 664,620, measured on the earlier single split, whose test block held one row more than the ten drawn here — whose denominators cancel to below 2×10^{-13}; every value returned is finite (the protocol's zero-fill for non-finite inversions is never triggered by the true components), and the floor is invisible in component errors. This floor is paid by every model. The ill-conditioning is continuous rather than confined to those entries: for the strongest classical model of the earlier single-split run, 7.4% of test entries land more than 0.5 away from the true reflectance, and 95% of those lie at bands whose true transmitted flux is below one percent of its median value. Reflectance is therefore summarized in what follows by its median absolute error, which the absorption windows cannot reach; outside the cancellation tail a perfect emulator scores zero to within 10^{-8}.

3 Models

All models are trained per component and evaluated in physical units after undoing the standardization; unless stated otherwise they regress the 64 leading principal coefficients of the standardized component, a reduction lossless at the 10^{-6} level (Section 2). Every hyperparameter named below is selected on the validation set of the seed at hand. Everything runs on CPU.

Reference family. Cubic polynomial ridge regression, acting on all monomials of degree at most three in the six standardized inputs, with the ridge weight selected from $\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. It is the benchmark of smoothness: a map that a cubic fits to two percent is a smooth map.

Exact kernel ridge regression. Three constructions, all Matérn kernels on the standardized inputs with the length scale expressed as a multiple of the median pairwise distance of the fit set. The first is the construction of the earlier study, kept for comparison: a Matérn-5/2 kernel fitted on a uniformly drawn subset of 4,000 training points, one eigendecomposition per length scale, multiplier in $\{1, 2\}$, nugget in $\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. The second is the exact solve on every training row: smoothness $\nu \in \{3/2, 5/2\}$, multiplier in $\{0.5, 1, 2, 4\}$ and the same nugget grid, tuned on validation from a 6,000-row training subsample and the winner refit on all 18,884 rows by one Cholesky factorization. The third gives each input its own length scale: starting from the isotropic winner, a coordinate search over per-input multipliers in $\{1/4, 1/2, 1, 2, 4\}$, two sweeps, scored on validation, after which the smoothness, scale and nugget are re-tuned at the found metric and the solve is refit on every row. The input relevance measured in Section 2 spans more than an order of magnitude across the six coordinates, which is what the third construction is for.

Fully connected networks. Three ELU hidden layers of 512 units per component, trained on the 285 standardized bands with AdamW at learning rate 10^{-3}, cosine decay, batch size 1024, weight decay 10^{-6}, up to 150 epochs with early stopping on validation loss (patience 25). Five networks are trained per component and seed from different initializations; the table reports the first and the mean of the five.

Kernel corrections of the neural mean. The construction of [6], with the exact stages now fitted on every training row. (i) *Residual correction*: an exact Matérn regression, on the standardized input space, of the network's training residuals in coefficient space, added back to the network, its smoothness, scale and nugget selected by the validation error of the corrected prediction; applied to the single network and to the five-network mean. (ii) *Kernel on features*: the same exact regression with the input metric replaced by the network's last hidden layer, standardized, so that the kernel's exactness is spent in coordinates the network has already straightened. This is the post-hoc counterpart of deep kernel learning [29], in which the representation and the kernel are trained together, and of the kernel-flow regularization of a network's inner layers [12]; here the representation is fixed first and the kernel is solved exactly on it. (iii) *Per-coordinate selection*: for each of the 64 output coefficients independently, pick on validation among the exact kernel, the network, the ensemble, the two corrected predictors and the feature kernel, and assemble the test prediction from the winners. (iv) *Convex stack*: per component, a convex combination of the same six heads with weights fitted on the validation

model	component rel. L^2 [%] Y_1	Y_2	Y_3	Y_4	radiance rel. L^2 [%]	reflectance med. $\lvert\hat{\rho}-\rho\rvert$ [%]
Cubic ridge	1.16±0.02	2.18±0.04	1.11±0.03	2.98±0.06	0.760±0.011	0.471±0.010
Matérn KRR, 4000-point fit	0.17±0.01	0.61±0.04	0.29±0.02	2.44±0.87	0.332±0.029	0.178±0.030
Matérn KRR, exact on all rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
Matérn KRR, per-input scales	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
FC-DNN (3×512)	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
Five-network mean	0.37±0.01	0.67±0.03	0.28±0.01	2.51±0.09	0.251±0.006	0.163±0.003
DNN + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
Ensemble + residual KRR	0.08±0.00	0.27±0.03	0.13±0.00	2.42±0.17	0.123±0.007	0.080±0.008
Kernel on network features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
Per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
Convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003

Table 1: Test errors under the protocol of Section 2, mean ± standard deviation over the ten splits. Component columns are mean relative L^2 errors; the radiance column evaluates (1) at $\rho = 0.7$; the last column is the median absolute reflectance error of the inversion (2), in reflectance percentage points, under the zero-fill evaluation of Section 2. The median is used because the sensitivity of (2) diverges continuously as the transmitted flux vanishes, so mean-square and tail-percentile summaries are dominated, for every model alike, by near-opaque entries; the same cancellation tail puts an RMSE floor of about half a percentage point under a perfect emulator (Section 2). The tail summaries are stored with the per-seed records.

points in the physical relative metric, in the manner of stacked generalization [27] and stacked regression [28]. Section 6 examines what the right weighting rule is for pools of this kind.

What is not in the table. Learning curves at nested training sizes, refits at other output ranks, and a wider and longer network were planned for the same seeds; the campaign that carried them stopped before they completed, and no number from them is used here.

4 Results

Table 1 collects the test errors. Every entry is a mean over the ten splits with its standard deviation, and the spreads are small enough that most of the orderings discussed below hold seed by seed; where they do not, the text says so.

The scale of the problem. The cubic ridge reaches 1.86% mean component error and 0.76% on the radiance: the map is smooth, and a polynomial in six variables recovers it to two percent. The earlier construction of the kernel regression, a Matérn-5/2 on a 4,000-point subset, divides both numbers by more than two at seven of the ten splits (0.74% and 0.32%); at the other three its albedo fit collapses (3.3–4.3% against 1.9%), which is what a fit capped at a random subset can do, and its ten-split mean (0.88%, 0.33%) carries that fragility.

Exact kernels. Fitting the same kernel on every training row, one Cholesky factorization of the 18,884 × 18,884 Gram matrix per component, lowers the radiance error to 0.24% and the path-radiance error by half (0.08% against 0.17%), and it has no collapsing split (2.3–2.9% on the albedo at every seed), but where the capped fit is sound it is worse on the albedo (2.6% against 1.9%), so the mean component error moves the wrong way at those splits (0.83% against 0.74%). The albedo is the component whose principal spectrum decays slowest and whose values sit within a few percent of zero over much of the grid, and it is also the one where the transfer of a scale and nugget tuned on a 6,000-row subsample to the full solve is least reliable. Giving each input its own length scale removes most of this: the per-input kernel reaches 0.63% on the components and 0.095% on the radiance, with the best reflectance column of the table (0.036 percentage points), and its albedo column (2.2%) sits between the two isotropic solves. The learned metric is the physics of the components: for the two flux components and the spherical albedo the coordinate search gives the relative azimuth a length scale sixteen times longer than the others — at every one of the ten seeds for the two flux components, and at nine of the ten for the albedo, whose search leaves the azimuth unscaled at the tenth — and the view zenith two to sixteen times longer, while for the path radiance, which does depend on the viewing geometry, the metric stays nearly isotropic. Transmittances and the spherical albedo do not depend on azimuth, and the kernel finds that out from the validation rows alone. On this table the per-input exact kernel and the kernel on network features (below) are, within the seed-to-seed spread, the strongest single models.

Networks. The fully connected network lands at 1.22% and 0.39%; the mean of five independently initialized networks at 0.96% and 0.25%, a third off the radiance error for five times the training cost, the usual decorrelation gain. Neither reaches the isotropic exact kernel on any of the first three components, and the network's advantage on the albedo that the single-split study reported does not survive replication: on Y_4 the five-network mean (2.5%) is behind the 4,000-point kernel wherever that kernel's fit is sound (1.9% at seven of the ten seeds) and behind the exact kernel with per-input scales at every seed.

Kernel corrections of the neural mean. An exact Matérn regression of the network's residuals on the standardized inputs takes the single network from 0.39% to 0.14% on the radiance and the five-network mean from 0.25% to 0.12%; the kernel fitted in the network's last-layer features (DKR) reaches 0.61% and 0.098%, indistinguishable from the per-input kernel on the components and on the radiance and a little behind it on the reflectance median. The correction therefore does exactly what it did on the single split — it hands the network the kernel's accuracy on the smooth components while keeping the network's own representation — but with the kernel now allowed its own input metric the corrected network is no longer the lowest row: the ordering by radiance error is stack, per-input kernel, DKR, corrected ensemble, corrected network, with the first four within 0.04 percentage points of one another.

Selection and stacking. The per-coordinate selection among the six heads, made on validation, does not transfer: its test radiance error (0.19%) is worse than four of the heads it selects among, and its spread over seeds (± 0.02) is among the largest in the table. The per-component convex stack, with weights fitted on the validation rows in the physical relative metric, is the best aggregate row (0.57% components, 0.087% radiance, 0.048 reflectance points) and wins the path radiance and the albedo columns outright; its gain over the best single model is eight percent on the radiance and on the components, of the order of one standard deviation of either. The paired view is sharper than the marginal one: split by split, the stack beats the per-input kernel on the components in nine of the ten splits (mean difference 0.06 points, spread 0.05) and beats DKR in all ten (0.047 ± 0.006), while the per-input kernel and DKR trade places (six of ten). The stack is the best row, and the residual correlations say why it can be: on the path radiance the residuals of the per-input kernel and of the five-network mean are correlated at 0.23 on average, below the ratio of their errors (0.33) at every one of the ten splits, which is the classical admission condition for a combination to improve on its better member [30], in the form used in [6]; the kernel and the corrected ensemble, at 0.63, are largely two views of the same predictor. The heads are complementary in the sense that matters for a stack — the kernel is best on the smooth components, the corrected networks on the albedo — and a per-coordinate winner-take-all rule is too brittle a way of using that complementarity when the validation and test blocks are drawn from the same table but the picks are made on 2,098 rows.

What changed from the single split. The earlier version of this study reported the corrected network as the lowest row, at 0.25% radiance error, and the exact kernel on the raw state at 0.41%; both numbers came from a single split whose test block had been consulted during exploration and from a kernel fitted on 4,000 points with one length scale. Under the fresh-seed protocol the corrected network reads 0.14% and the exact kernel with per-input scales 0.095%. The qualitative claims that survive are the smoothness of the map, the complementarity of the heads across components, and the mechanism of the residual correction; the claim that the corrected network is the best model on this table does not.

The albedo. Every family is worst on Y_4 by an order of magnitude, and the per-band errors show why: the spherical albedo's values in the short-wave infrared are close to zero, so relative error is large for every model there, and the component is hard for reasons of signal-to-noise rather than of model class. The stack's 1.94% is the best number any construction reaches at this training size.

5 A second emulation problem: OCO-2

The same construction was applied, without structural change, to the OCO-2 radiative-transfer emulation problem of Lamminpää et al. [5]: per spectral band (O_2, weak CO_2, strong CO_2), a reduced atmospheric state of twenty or twenty-four coordinates maps to forty reduced radiance coefficients, with the release's 20,000 training and 2,000 test points, and errors reported both on the standardized reduced coefficients and on the reconstructed monochromatic radiance. The protocol is the one of Section 2: ten splits of the training block into 18,000 training and 2,000 validation rows, every choice on validation, one test read per family, and the fixed test block scored identically for every row. The reference is the release's own kernel-flow emulator, whose stored test predictions are rescored under the same two metrics on the same 2,000 points, so that the reference row and every other row are produced by one evaluation routine on one set of cases. An earlier version of this study quoted reference figures that the stored predictions do not reproduce under any convention we tried; the rows below compare against the rescored predictions only.

This problem sits at the opposite end of the structural spectrum from the EMIT table. The input is higher-dimensional, the map is rougher, and a kernel on the state trails the network by an order of magnitude on the reduced coefficients: the isotropic Matérn reaches 41%, 59% and 43% on the three bands and the relevance-scaled metric 30%, 52% and 29%, against 4.4%, 16.4% and 8.2% for the network. Representation carries the problem, so the construction to test

model	reduced coefficients rel. L^2 [%]			radiance rel. L^2 [%]		
	O_2	weak CO_2	strong CO_2	O_2	weak CO_2	strong CO_2
reference emulator, stored predictions	16.89	24.06	16.14	0.0448	0.0599	0.1147
trained on the reduced coefficients						
network	4.43±0.05	16.44±0.06	8.23±0.02	0.1949±0.0139	0.2260±0.0120	0.1990±0.0147
ridge readout on its features	4.71±0.07	16.51±0.06	8.33±0.03	0.2276±0.0143	0.2523±0.0167	0.2359±0.0318
Matérn kernel on its features	4.11±0.05	16.28±0.06	8.08±0.01	0.0793±0.0072	0.1159±0.0065	0.1093±0.0065
trained on the weighted coefficients						
network	49.33±0.40	62.57±0.56	36.46±0.65	0.0439±0.0031	0.0738±0.0057	0.0751±0.0025
ridge readout on its features	45.71±0.61	61.32±0.43	40.10±0.91	0.0521±0.0043	0.0842±0.0067	0.0866±0.0039
Matérn kernel on its features	21.72±0.52	48.67±0.45	15.47±0.47	0.0316±0.0019	0.0533±0.0054	0.0631±0.0037
trained on the reconstructed radiance						
network	59.05±0.87	67.49±0.41	46.76±0.79	0.0575±0.0057	0.0649±0.0127	0.0927±0.0084
ridge readout on its features	45.59±1.13	61.01±0.90	40.13±1.07	0.0602±0.0063	0.0690±0.0115	0.0951±0.0078
Matérn kernel on its features	20.47±0.61	48.23±0.31	15.49±0.66	0.0379±0.0037	0.0486±0.0053	0.0766±0.0051
per-coordinate selection over the six heads	4.12±0.05	16.28±0.06	8.08±0.01	0.0291±0.0020	0.0526±0.0053	0.0586±0.0039

Table 2: OCO-2 test errors by training loss and readout, mean ± standard deviation over ten splits, on the release's 2,000 test points. Each block is one network trained under the named loss, read out by its own head, by a ridge regression refitted on its frozen last-layer features, and by an exact Matérn regression on the same features. The per-coordinate selection chooses among the six network and kernel rows, coefficient by coefficient, on validation. The reference row is the release's kernel-flow emulator rescored on the same points; it carries no spread because it does not depend on the split.

here is the one that spends the kernel's exactness inside a learned representation rather than on the state.

5.1 What the kernel adds to a representation

Table 2 separates that question from two others it is easily confused with. Each of three networks is trained under a different loss — the per-sample relative error on the reduced coefficients, the same error under the weights the reconstruction applies to each coefficient, and the relative error of the reconstructed radiance itself — and each is then read out in three ways: the network's own output head, a ridge regression refitted on its frozen last-layer features, and an exact Matérn regression on those same features. The ridge row is the control. If the kernel's gain over the network were a matter of refitting a readout on a good representation, the ridge row would match the kernel row; if the representation were doing all the work, the ridge row would match the network.

Neither happens. The kernel head beats the ridge readout on the reduced coefficients at all ten splits, on every band and under every loss, by 0.60 ± 0.07 points on O_2 under the reduced loss and by twenty-four points under the weighted loss; the paired bootstrap interval of that difference lies entirely below zero at every split. The ridge readout, meanwhile, is worse than the network it reads out from wherever the network was trained for the metric being scored — 4.71% against 4.43% on O_2, at all ten splits — and better than it in five of the six blocks where the network was trained for a different metric, which is what a refitted linear readout should do. The exception is the weighted-loss network on strong CO_2, where the readout is worse at all ten splits. What the kernel head adds is therefore not attributable to refitting a readout on a good representation, and not to the representation alone.

5.2 Which metric the emulator is trained in

The three losses divide the two metrics between them, and neither metric is a proxy for the other. Trained on the reduced coefficients, the kernel head reaches 4.11%, 16.28% and 8.08% there and 0.079%, 0.116% and 0.109% on the radiance; trained on the weighted coefficients it reaches 0.032%, 0.053% and 0.063% on the radiance and two to five times worse on the coefficients (5.3 times on O_2, 3.0 on weak CO_2, 1.9 on strong CO_2). The reconstruction basis is orthogonal — the projection rows satisfy $PP^\top = I$ to machine precision on all three bands — so the squared radiance error is a weighted sum of per-coefficient squared errors whose weights span several decades and concentrate on the leading few coordinates, while the reduced metric weights all forty alike after standardization. A model trained on the coefficients spends its accuracy on the tail, which decides that metric; a model trained through the reconstruction spends it on the head.

Because the basis is orthogonal, a per-coefficient choice can take both. The selection made on validation over the six heads assigns the leading coefficient to the kernel head of the weighted-coefficient network at every split on every band, and every remaining coefficient to the kernel head of the coefficient-trained network, with five of the four hundred slots going to that network itself on O_2 and on weak CO_2 and none on strong CO_2. The radiance-trained family, whose loss is the reconstruction itself, is never selected at any of the twelve hundred slots: the weighting the reconstruction induces is a better training signal here than the reconstruction error. One coefficient of forty decides the radiance metric. The result is 4.12%, 16.28% and 8.08% on the coefficients and 0.029%, 0.053% and 0.059% on the radiance, against the stored emulator's 16.89%, 24.06% and 16.14% and 0.045%, 0.060% and 0.115%: lower on both metrics on all three bands. Paired on the same test points, the selection beats the emulator on the coefficients at all ten splits on every band, and on the radiance at all ten on O_2 and strong CO_2 and at nine of ten on weak CO_2; the paired bootstrap interval of the difference lies entirely below zero at every split except two on weak CO_2.

The earlier single-split version of this study reached neither metric and gave a reason: its member set varied architecture and kernelization within one training objective, and no per-coordinate selection can produce an error profile that none of its members has. That reading predicted the present result, and the enlargement it called for — objectives spanning the reported metrics — is what Table 2 supplies. The productive axis on this problem is the loss, not the architecture.

5.3 Ensembles

A second campaign on the same ten splits trains three networks per band under the reduced loss and combines them (Table 3). Their mean improves the single network slightly; an exact Matérn regression on the members' concatenated features is the best of the retrained rows on both metrics and on every band, at 3.81%, 16.11% and 7.94% on the coefficients and 0.053%, 0.070% and 0.075% on the radiance, and beats both the single-member kernel head and the plain three-network mean at all ten splits under both metrics. This pool is not the one that overtakes the reference on both metrics: with every member trained on the coefficients, the stored emulator is still ahead on the radiance metric on O_2 and weak CO_2, which is the point Section 5.2 makes with the losses varied. Averaging the members' own kernel heads instead of fitting one kernel on their joint features is slightly behind everywhere, so part of the gain belongs to the joint feature space and not to the averaging alone. The per-coordinate selection within

model	reduced coefficients rel. L^2 [%]			radiance rel. L^2 [%]		
	O_2	weak CO_2	strong CO_2	O_2	weak CO_2	strong CO_2
reference emulator, stored predictions	16.89	24.06	16.14	0.0448	0.0599	0.1147
Matérn KRR on the state, isotropic	40.57±0.10	58.57±0.14	43.40±3.36	0.2331±0.0034	0.4391±0.0072	0.7070±0.3377
Matérn KRR on the state, scaled metric	29.98±0.15	52.39±0.08	28.66±0.14	0.2040±0.0865	0.2230±0.0011	0.5991±0.0122
network	4.43±0.07	16.44±0.08	8.22±0.03	0.1796±0.0155	0.2189±0.0128	0.1925±0.0151
DKR: Matérn KRR on network features	4.11±0.07	16.27±0.06	8.08±0.02	0.0737±0.0063	0.1143±0.0056	0.1130±0.0107
mean of three networks	4.04±0.05	16.23±0.03	8.04±0.02	0.1179±0.0114	0.1342±0.0067	0.1428±0.0092
DKR on concatenated features (3)	3.81±0.04	16.11±0.03	7.94±0.02	0.0528±0.0066	0.0704±0.0046	0.0746±0.0055
mean of three DKR heads	3.85±0.04	16.13±0.02	7.96±0.02	0.0538±0.0058	0.0762±0.0040	0.0849±0.0062
per-coordinate selection	3.83±0.06	16.11±0.03	7.95±0.02	0.0528±0.0066	0.0704±0.0046	0.0746±0.0055

Table 3: OCO-2 test errors for the three-member campaign, mean ± standard deviation over the same ten splits, all networks trained under the reduced-coefficient loss. The reference row is shared with Table 2 and is recomputed here from this campaign's own records; the two kernels on the state appear only here, and are the figures Section 5 quotes for a kernel fitted on the raw state.

this pool has nothing to choose: it assigns the concatenated-feature kernel to 384, 390 and 397 of the four hundred coordinate slots, matching it on the radiance metric and on weak CO_2 and trailing it by 0.02 points on the O_2 and strong CO_2 coefficient columns.

The two campaigns share their splits but trained their networks independently, which makes the rows they have in common a replication. Taking the larger of the two rows on each band, the single network and its kernel head agree between the campaigns to 0.006, 0.008 and 0.010 points in the mean, with a largest per-split difference of 0.26 points.

5.4 Two constructions that do not transfer

The residual correction, the EMIT table's second-best construction, adds nothing on any OCO-2 band, as [6] also found on this problem. On a rough map the residual of a trained network has no smooth part left for an exact kernel on the state to fit, and the exactness is better spent inside the representation. The coordinate search that gives a kernel on the state its own length scales, which is the strongest such construction on the EMIT table, is also not the strongest here: on a single lane run before the campaign ended (O_2, one split, the same 18,000 rows), per-input scales fitted by empirical Bayes take the isotropic kernel from 40.6% to 17.3% on the coefficients and from 0.235% to 0.113% on the radiance, against 27–28% for a kernel-flow metric [31], for a Mahalanobis metric under the same flow, and for the gradient outer product of the recursive feature machine [15], and 38.6% for an additive kernel. One lane at one split is a direction and not a result; what it indicates is that a learned metric closes part of the gap on the state, and that none of the four metrics tried brings a kernel on the state within a factor of four of a kernel on the network's features.

5.5 Reevaluation on the retrieval records

The rows of this subsection come from the earlier single-split version of the OCO-2 study: three families of networks trained per band under different losses, their per-coordinate validation selection, and a larger-budget relative-loss ensemble, all evaluated once. They are retained because the finding about the records themselves does not depend on the protocol, and they are reported as what they are.

The reference's data release carries 976 retrieval records, each with an atmospheric state, the full-physics monochromatic radiance at that state, the rank-40 reconstruction of that radiance in the release's basis, and the prediction of the release's Gaussian-process emulator at the same forty coefficients. A direct identity check finds that every one of the 976 reduced states matches a stored test input within 6.67×10^{-16}; the records map to 974 distinct test rows, and the nearest test-row index is the same in all three bands. The records are thus not disjoint from the stored test set and cannot serve as a new or independent evaluation pool. They do provide a different target — the full-physics radiance attached to those states. The state reduction was replayed independently from the release's two dimensional-reduction files and the `x_in` and `geo_in` fields of all 976 records; the replay code and its digests are archived with the per-run records. That replay establishes the input identity stated above; it does not replay the emulator-error rows. Those rows are retained from the recorded aggregate results. This snapshot does not contain the raw arrays, trained weights, or prediction dumps needed to retrain the three network families or reevaluate their individual predictions.

Table 4 reports the results on this test-associated reevaluation. The ordering of the training objectives under the full-physics target is similar: the relative-loss ensembles have the lowest error among the retrained network families on every band, ahead of mean-squared training, with the radiance-weighted nets far behind. The reference's per-coordinate Gaussian processes stay 2.1 to 5.3 times ahead of the best network family on these records, while the rank-40 floor sits one to two orders of magnitude below every emulator, so the reduction is nowhere the binding constraint. And the fragility of thin validation picks is a measured phenomenon under this alternate target: the selection carries the weighted head's leading-coordinate pick on O_2 and weak CO_2, and across the 976 records that one coordinate leaves the selection a factor 2.4 behind its own best member, while on strong CO_2, where validation chose the relative-loss head at every coordinate, selection and member coincide exactly. The larger-budget relative-loss ensembles add a final datum (first row of Table 4): they improve strong CO_2 on the records by forty percent, leave O_2 essentially unchanged, and are worse on weak CO_2 than the smaller ensembles they dominate under the stored test metric. This reversal shows target sensitivity at the same test-associated inputs; because the inputs overlap the test set, it is not evidence of generalization to a third distribution.

6 Weighting the heads: a minimum-variance stack

The convex stack was the best row of the EMIT table and the per-coordinate selection the most fragile, and both are linear combinations of the same heads fitted on the same validation rows. This section asks what the right combination rule is, on the member predictions already produced by the two campaigns and without retraining anything. The pool on EMIT is the 23 heads available per split and component — the kernels, the networks, the corrections, the feature kernels, and the learned-kernel and regularized-network variants of the campaign — at the ten splits and four components, forty cells; the pool on OCO-2 is the 18 heads per split and band, thirty cells. The weights of this section are not fitted on the validation rows the tables above use. Each split's test block is divided instead, 932 calibration rows on EMIT and 800 on OCO-2 against the remainder, the weights are fitted on the calibration part and every number below is scored on the rest. The comparison between rules is therefore internally consistent, but the levels are not comparable with the test errors of Tables 1 and 3, and no rule studied here is a rule the earlier tables could have used. Every rule is scored in the physical relative metric of

	monochromatic radiance rel. L^2 [%], mean over 976 records		
	O_2	weak CO_2	strong CO_2
relative-loss network, larger budget	0.2430	0.3846	0.2371
relative-loss network	0.2325	0.3155	0.3964
mean-squared-loss network	0.3792	0.5909	0.4950
radiance-weighted network	0.6055	0.7929	1.0294
per-coordinate validation selection	0.5489	0.7732	0.3964
reference GP, 40 coefficients	0.0453	0.0594	0.1148
rank-40 projection floor	0.0005	0.0009	0.0052

Table 4: Mean relative L^2 error on the monochromatic radiance over the 976 retrieval records of the reference's release, whose reduced inputs coincide with 974 distinct stored test rows. This is a reevaluation against the records' full-physics radiances, not an independent holdout. Rows give the three retrained network families, the per-coordinate selection fixed on the table's validation split, the release's own Gaussian-process emulator at forty coefficients, and the rank-40 projection floor. The identity mapping is independently replayed; the emulator-error rows are recorded aggregates and are not numerically reproduced by this snapshot.

the corresponding table, and every number below is the change in that error relative to a named baseline, averaged over cells, with the number of cells on which the rule is better.

What was tried and what was wrong with it. The obvious rules bracket the problem. The baseline every other rule is measured against is the strongest heuristic the pool already had: a per-output ridge whose penalty is chosen by leave-one-out, shrunk toward the pool's global convex weights by an amount chosen with five-fold cross-validation on the calibration rows. Against it, equal weights lose 151% on EMIT and 85% on OCO-2, because the pools contain weak members; the best single head chosen on calibration loses 20% and 2.3%; and per-output least squares loses 5.1% and 0.9%. The rules random-matrix theory suggests first were applied, in the pool as we found it, to the covariance of the member *predictions*: clipping at the Marchenko–Pastur edge [16, 32], a rotationally invariant estimator [17], and a linear shrinkage [18]. They fail badly, and for a structural reason: the leading eigenvalue of the prediction Gram is the shared signal, two orders of magnitude above the rest, while what separates one weighting from another lives in the small eigenvalues these estimators discard.

The object the theory is about is the covariance of the member *errors* in the reported metric. Cleaning that matrix instead is not by itself enough: the spiked estimator loses 11.6% to the baseline on EMIT and an unconstrained rotationally invariant estimator 30.9%, while the same estimator combined with the pool's pooled shrink gains 2.2%. The error covariance is nearly singular — the members are close to collinear, with an effective dimension of two or three and condition numbers of 10^6 to 10^{17}, so the ratio of members to calibration rows understates the estimation problem by an order of magnitude — and its spectrum is dense rather than spiked, eleven of twenty-three eigenvalues lying above the Marchenko–Pastur edge on a typical EMIT cell, which is why the spiked model is the worst of them. The near-zero eigenvalues are genuine rank deficiency from near-duplicate members and not sampling noise, a separate diagnostic.

The rule that survives. Writing the stack as a minimum-variance portfolio on the error covariance, in the reported metric, with an intercept and with the weights nonnegative and summing to one — gross exposure one in the sense of [22] — gives a rule with no cross-validated quantity and no tuned constant. On EMIT it beats the pool's best heuristic by 4.6% on average (33 of the 40 cells; 8.8% against per-output least squares, and 9.3% on the radiance metric at all ten splits), and the gain is about 5% at every smaller calibration size tried (60, 120 and 240 rows), so it is not bought with the full calibration block. On OCO-2 the same rule loses 1.3% on all thirty cells. The two corpora differ the way the mechanism predicts: EMIT has 285 outputs, heterogeneous members and a large mismatch between the members' training losses and the scored metric, and there the constraint removes estimation variance; OCO-2 has forty outputs, members within a factor of six of one another, and almost no mismatch, and there pooling the weights across outputs is the regularizer that matters. A ladder of seventy cells over the covariance cleaners — the raw sample covariance, the rotationally invariant estimator at several floors, the linear and the analytical nonlinear shrinkage of Ledoit and Wolf [18, 19], and the nonparametric eigenvalue-regularized estimator [20] — shows that inside the nonnegativity constraint every consistent cleaner ties to within a percent at full calibration. What loses is corpus-dependent and smaller than that ladder's spread suggests: on EMIT a floor at one σ^2 and the linear shrinkage each lose about 6% and a floor at $0.2\sigma^2$ about 1%, while on OCO-2 the same cleaners lose at most 1% and the linear shrinkage does not lose at all. The constraint, not the cleaning, is the estimator. This is the mechanism Jagannathan and Ma identified for portfolio weights [21]: a nonnegativity constraint is itself a shrinkage of the covariance, and the entrywise bound of Fan, Zhang and Yu [22] says that the excess risk of a constrained plug-in is at most twice the largest entrywise error of the covariance estimate, with neither the member-to-row ratio nor the condition number in it. Measured on every output, the bound holds and is loose by one to three orders of magnitude; what it delivers is finiteness and the right dependence. The same reading explains where the gain comes from: the pairs-bootstrap variance of the least-squares weights at the rows the stack is scored on tracks the cell's gain inside every output group (rank correlation 0.69 across the forty EMIT cells with the group effect removed, and 0.74 on a fresh split). The association is within-split: the same predictor taken from one split and applied to another falls to 0.17, so it describes which cells gain on the rows at hand rather than which cells gain in general. Separately, removing the tenth of rows with the largest gains and losses leaves 87% of the EMIT gain, so the constraint pays broadly across the evaluation rows rather than on a few.

Choosing without labels. The corpus dependence is removed by two label-free choices. A five-fold cross-validation on the calibration rows alone, choosing per cell among the constrained stack, its shrunk form, the sum-to-one rotationally invariant stack and the pool's own heuristic, lands 6.0% above the pool's best on EMIT and level with it on OCO-2, within 0.4% of the oracle of the four. A blend of the constrained and least-squares stacks, with the mixing weight chosen by cross-validated calibration error in the score's own metric over a three-vertex grid that includes the pool's heuristic, keeps the constrained stack where it pays and beats both endpoints where neither does. On a split neither the estimator nor the criterion had seen it reads 5.3% above the pool's heuristic on EMIT and 0.10% above it on OCO-2, with no hand tuning; the earlier squared-loss form of the same criterion reads 4.3% and 0.06% on that split, which is why the criterion is written in the score's own metric. A transductive term that used the members' predictions at the scored rows was tried in several forms and never earned its place; the shipped rule has no bootstrap and nothing transductive. Below a few hundred calibration rows the blend loses its footing and the constrained stack alone is the better choice.

What this changes in the tables. The EMIT stack row of Table 1 is the convex stack fitted per component on the full validation block, which is the constrained rule at its simplest; the numbers above say that this row is the right kind of combination for that pool, that no covariance cleaning would move it, and that a least-squares stack in its place would be worse by about nine percent. On OCO-2 the members are homogeneous, the best and worst differing by a factor of six, and every rule in the ladder lands within two percent of every other; the constrained rule is not the one to use there, and this section does not score the per-coordinate selection that Table 3 reports, so it says nothing about it.

7 The reading on other corpora

The two problems above stand at opposite ends of one axis: a smooth, six-dimensional, low-rank table where an exact kernel with a learned metric is the strongest single model, and a rougher, twenty-dimensional one where the kernel is useful only inside a learned representation. If that axis is the right way to read the comparison, it should order other emulation problems too. Table 5 applies the same family set to ten further configurations, with no tuning per corpus beyond what each family selects on validation.

The corpora are the advection equation with discontinuous initial data from the operator suite of [23], which is also the source of the structural-mechanics problem of [6]; Burgers' equation at three viscosities from PDEBench [24]; three one-step maps from The Well [25] — a two-dimensional turbulent radiative layer, an active-matter system, and a Helmholtz staircase; the ClimSim atmospheric physics emulation task [26]; and the paired correction-coefficient corpus of the physics-guided radiative-transfer work of [4], which is a third radiative-transfer emulation problem and the largest here at 487,790 rows. That last corpus is run in both of its configurations: from the atmospheric state alone, and with the low-fidelity 6S coefficients supplied as additional inputs, which is a different and easier problem. Inputs and outputs are reduced by principal components as on the EMIT table, and the families are those of Section 3.

What each row averages differs and is stated in the table. Advection, Burgers, ClimSim and the two correction-coefficient configurations have three to five random splits. The Well's releases carry their own train, validation and test files, which are used unchanged, so the three turbulent-radiative-layer runs share one split and differ only in network initialisation; their spread, of order 0.01 points, measures initialisation and not sampling. The active-matter and Helmholtz-staircase rows are single runs. On ClimSim, the two correction-coefficient configurations and the Helmholtz staircase the kernel solves are capped rather than exact, because an exact Gram matrix at those sample sizes does not fit in memory; those kernel entries are therefore an upper bound on the error an exact solve would give.

Three things in the table repeat what the two main problems show. First, the ordering between families follows the structure rather than the corpus. Where the map is smooth and the sample small — Burgers at $\nu = 0.1$ — the exact kernel is at 2.81% against the network's 20.34%, a factor of seven, of the same kind as the five- to six-fold gap between the per-input kernel and the network on the EMIT table's smooth components; as the viscosity falls and the solutions roughen the gap closes to 23.40% against 36.65% and then to 31.66% against 39.98%. Where the sample is large and the map rough — ClimSim, and the correction-coefficient corpus at nearly half a million rows — the network leads instead, 25.10% against 27.20% and 3.75% against 62.93%, which is the OCO-2 regime. Where the operator is linear, on advection, a cubic ridge regression is the lowest row and every other family lands within 0.4 points of it, as it

corpus	n	d	q	runs	cubic ridge	KRR	KRR, learned metric	network	network ensemble	network + residual KRR	kernel on features	convex stack
Advection	19,000	200	200	5	11.29±0.04	11.36±0.04	11.36±0.04	11.69±0.07	11.50±0.05	11.50±0.05	11.64±0.07	11.36±0.04
Burgers $\nu = 0.1$	7,200	256	128	3	13.89±0.53	2.81±0.28	2.90±0.33	20.34±0.61	19.28±0.09	2.78±0.29	4.92±0.30	2.78±0.29
Burgers $\nu = 0.01$	7,200	256	128	3	33.68±0.28	23.40±0.54	23.58±0.50	36.65±1.95	35.31±0.70	23.12±0.56	29.20±0.59	23.01±0.54
Burgers $\nu = 0.001$	7,200	256	128	3	38.38±0.12	31.66±0.16	33.23±1.73	39.98±0.65	38.89±0.11	31.82±0.11	35.59±0.48	31.34±0.14
Turbulent radiative layer	7,200	256	256	3	31.73	30.31	30.26±0.01	32.17±0.02	31.58±0.02	30.40±0.01	31.08±0.01	30.12
Active matter	14,000	256	256	1	37.24	35.81	35.18	37.31	36.93	35.77	36.22	35.05
Helmholtz staircase	20,384	256	256	1	0.00	0.26	0.07	0.40	0.24	0.23	0.45	0.07
ClimSim	100,000	124	128	5	32.24±0.06	27.20±0.05	26.31±0.04	25.10±0.18	24.43±0.17	24.33±0.13	24.71±0.25	24.25±0.17
RT corrections, state inputs	487,773–487,823	8	3	5	71.54±0.59	62.93±0.51	3.97±0.42	3.75±0.25	3.24±0.11	3.13±0.05	3.52±0.07	3.09±0.05
RT corrections, multi-fidelity inputs	487,773–487,823	11	3	5	33.83±0.26	15.76±1.98	4.60±0.31	2.79±0.14	2.40±0.11	2.34±0.09	2.80±0.09	2.32±0.08

Table 5: Mean relative L^2 test error [%] on ten further emulation configurations, mean $\pm$ standard deviation over the runs available for each. n is the number of examples, d the reduced input dimension, q the reduced output dimension. The runs are random splits except for the turbulent radiative layer, whose three runs share The Well's own split and differ in network initialisation, and the two single-run rows. The family set, the principal-component reduction and the validation-based selection are common to every row; the network budget is not, and is 200 epochs at width 512 except on ClimSim (60 at 512) and the two correction-coefficient configurations (120 at 384). Kernel solves are exact except on ClimSim, the Helmholtz staircase and the two correction-coefficient rows.

should.

Second, a learned metric is what rescues a kernel on the state when the plain one fails, and it is worth most where the plain kernel is worst. On the correction-coefficient corpus from the state alone the isotropic kernel is at 62.93 ± 0.51 and a kernel-flow metric [31] takes it to 3.97 ± 0.42, a factor of sixteen in the mean, which brings a kernel on the state to within six percent of the network on a corpus where the plain kernel was seventeen times worse. With the low-fidelity coefficients supplied as inputs the same pair reads 15.76 and 4.60, a factor of three: the easier the problem is made for the plain kernel, the less the learned metric adds. This is the same lever as the per-input length scales on the EMIT table and the empirical-Bayes metric of Section 5.4, at a scale where its effect is unambiguous.

Third, the combination is at or near the best family everywhere, and the one place it is not tells the same story from the other side. The convex stack of Section 6 is the lowest row on eight of the ten configurations and the second lowest on the other two; the residual-corrected network is in the lowest two on five. The two exceptions are the linear problems: on advection the cubic ridge is lower than the stack by 0.07 points, and on the Helmholtz staircase, where a cubic ridge regression reaches 0.004% and is three orders of magnitude better than anything else in the pool, the stack reaches only 0.066%. A convex combination cannot recover a member's accuracy when that member is better than the rest by orders of magnitude and the weights are fitted on a finite calibration sample; the safe reading is that a combination is a good default when no family is known to dominate, and a poor one when the structure says a single family should.

Two limits on scope. These rows use one training budget per corpus and one architecture family, and several of these corpora have published results from methods tuned for them specifically; the comparison here is between families under a common protocol, not against the state of the art on each problem. And with three to five runs, and one on two rows, the spreads support the large orderings above and not fine distinctions between adjacent families.

8 Discussion

Three things changed when the study moved from one split to ten. The first is the standing of the exact kernel. Fitted on a 4,000-point subset with a single length scale it was a useful baseline; fitted on every training row with a length scale per input it is the best single model on the EMIT table, and its metric is interpretable: the coordinate search finds, from the validation rows alone, that the transmittances and the spherical albedo do not depend on the relative azimuth. The second is the standing of the corrected network. Its mechanism is unchanged — the exact regression of the residuals hands the network the kernel's accuracy on the smooth components while the network keeps its own representation — but with the kernel allowed its own metric the correction is one of four heads within a few hundredths of a percent of one another on the radiance, not the lowest row. The third is the way heads are combined. A per-coordinate selection made on 2,098 validation rows does not transfer to the test block, and its spread across splits is among the largest in the table; a convex stack per component, fitted in the reported metric, is the best row at every split but one. We read this as a statement about calibration data rather than about the heads: with a few thousand rows the right way to use complementary predictors is a low-dimensional combination, not a per-output choice.

The OCO-2 problem says the same thing from the opposite structure. There the kernel on the state is an order of magnitude behind the network, the residual correction adds nothing, and the construction that pays is the exact kernel inside the network's representation. The ridge-readout control settles what that construction is worth and why: the kernel head beats a linear readout refitted on the same frozen features at every split, on every band and under every loss, while the linear readout is worse than the network it reads out from. The gain is the kernel, not the refitting, and it is available to any trained network at the cost of one Cholesky factorization.

The other thing OCO-2 shows is that the training loss, not the architecture, is the axis a selection can exploit. Networks trained on the coefficients and networks trained through the reconstruction sit at opposite ends of the two reported metrics, and because the reconstruction basis is orthogonal a per-coefficient choice made on validation can hold both ends at once: one coefficient of forty, assigned to the radiance-trained head at every split on every band, is what separates a selection that loses the radiance metric from one that wins it. The single-split version of this study reached neither metric and attributed the failure to a member set that varied architecture within one objective; the present campaign was built to test that attribution and confirms it.

Section 6 closes the loop on the combination rules: the convex stack that wins the EMIT table is the constrained minimum-variance rule in its simplest form; the random-matrix cleaners that seemed the natural refinement act on the wrong matrix; the constraint is the estimator; and on a pool that one head dominates, no combination rule is worth more than a percent. Section 7 then puts the reading on seven further corpora, where the family ordering follows the structure as it does here, a learned metric rescues a kernel on the state exactly where the plain kernel fails worst, and a combination is the lowest row on six of the seven.

The albedo sets the floor. Every family is an order of magnitude worse on Y_4 than on the other components, the per-band anatomy places the error where the albedo is nearly zero, and neither more capacity nor a different family moves it at this training size. An emulator that is used through the inversion (2) feels this floor only through the cancellation tail discussed in Section 2.

On cost, the exact solve is one Cholesky factorization of an $18{,}884 \times 18{,}884$ matrix per component and one kernel row per query; the per-input search multiplies the tuning cost by the number of coordinates and leaves the solve unchanged. The networks cost more to train and less to query. The stack costs nothing beyond its members.

The comparisons that survive are the ones whose differences exceed the seed-to-seed spread.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author made substantial use of generative AI tools, and describes their role here. The software — the data pipeline, the model comparison, the diagnostics, and the figures — was written and executed by Claude Code (Anthropic) and Codex (OpenAI), which also prepared the first draft of this manuscript, working from the author's data and from the method and experimental recipe of [6]. The measurements of the data's structure, the identification of the ill-posed reflectance entries, and the reevaluation of Section 5.5 were produced in the same AI-assisted sessions and verified against the run summaries. The author reviewed the code, the numbers, and the text, and takes full responsibility for the content of the manuscript.

Data and code availability

The dataset (`data_EMIT_24k`) consists of tabulated radiative-transfer evaluations generated at the Jet Propulsion Laboratory; the OCO-2 data and the reference emulator's stored predictions are those of the companion paper's release; the corpora of Section 7 are public releases, cited there. The pipeline, the per-seed records behind every number reported here (one record per band, seed and family for OCO-2, one per seed for EMIT, and the seventy-cell stacking runs), and the exact configuration of each experiment are retained by the author and will accompany the published version of this manuscript; the method repository of [6] is at `https://github.com/yspennstate/neural-means-kernel-corrections`.

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